AP190 Practicum Portfolio

Submitted by:

MIEL GABRIEL LAGDAN
Bachelor of Science in Applied Physics
June 22, 2026 - July 31, 2026

Submitted to:

ERIC M. INOCENCIO, MSc., LPT
OJT Coordinator
Physics and Geology Unit
Department of Physical Sciences and Mathematics
University of the Philippines Manila

Contents

Course Description

AP 190 PRACTICUM GUIDELINES
FOR STUDENTS AND PROSPECTIVE SUPERVISOR

A. INTRODUCTION

What is the BS Applied Physics (Health Physics) program about?
The BS Applied Physics (Health Physics) program at the University of the Philippines Manila is a four-year undergraduate degree program that combines a strong foundation in physics with a focus on applications in the field of health and medical physics. The program provides students with a broad range of theoretical knowledge and practical skills related to radiation safety, medical imaging, radiation therapy, and nuclear medicine.

In particular, the program emphasizes the use of physics principles and techniques to understand the behavior of radiation in biological systems and the application of this understanding to the diagnosis and treatment of diseases. The program also emphasizes the importance of safety and regulatory compliance in the use of radiation in medical applications.
The program consists of a mix of classroom instruction, laboratory work, and hands-on experience through internships and research projects. Graduates of the program are prepared for careers in a range of fields, including medical physics, health physics, radiation safety, nuclear medicine, and radiation therapy, as well as for further study in graduate programs in physics, medical physics, or related fields.

What is the significance of the internship program?
The internship program is a crucial component of the BS Applied Physics (Health Physics) program at the University of the Philippines Manila. It provides students with an opportunity to apply their theoretical knowledge and skills in real-world settings, gain practical experience, and develop a deeper understanding of the field of health and medical physics.

Through internships, students can gain exposure to a range of professional settings, including hospitals, research laboratories, and regulatory agencies, and work alongside professionals in the field. This exposure provides them with valuable networking opportunities and can help them to identify potential career paths.

The internship program also enables students to develop important professional skills, such as communication, collaboration, and critical thinking, which are essential for success in the field of health and medical physics. Moreover, the internship program helps students to identify areas of interest within the field and to refine their career goals.

Overall, the internship program is an important aspect of the BS Applied Physics (Health Physics) program, as it helps to prepare students for successful careers in the field of health and medical physics and contributes to the development of a highly skilled and knowledgeable workforce in this critical area of healthcare.

B. OBJECTIVES

At the end of the program, the student should be able to deliver the following:
BS APPLIED PHYSICS (HEALTH PHYSICS) PROGRAM LEARNING OUTCOMES:

- Apply theoretical, computational, and experimental physics in a workplace;
- Conduct research related to health and medical physics that contribute to national development;
- Advance towards higher knowledge and experience requisite to assuming a leadership role in the health and medical physics profession;
- Collaborate with members of the health and medical physics community and representatives from related health and safety professions;
- Synthesize and effectively communicate scientific information in both English and Filipino; and
- Act in recognition of professional, social, and ethical responsibility.

with the following **AP 190 PRACTICUM COURSE LEARNING OUTCOMES (CLO):**

1. Demonstrate an understanding of theoretical knowledge on health physics by performing their duties as a student intern on their chosen institution for at least 200 hours
2. Exemplify professionalism and work ethical standards with different radiation personnel whether in a hospital or an industrial setting
3. Appreciate health physics as a developing field in the country
4. Create a portfolio summarizing the experience of a health physics intern in the field (hospital, research, or industrial setting)

C. General Guidelines

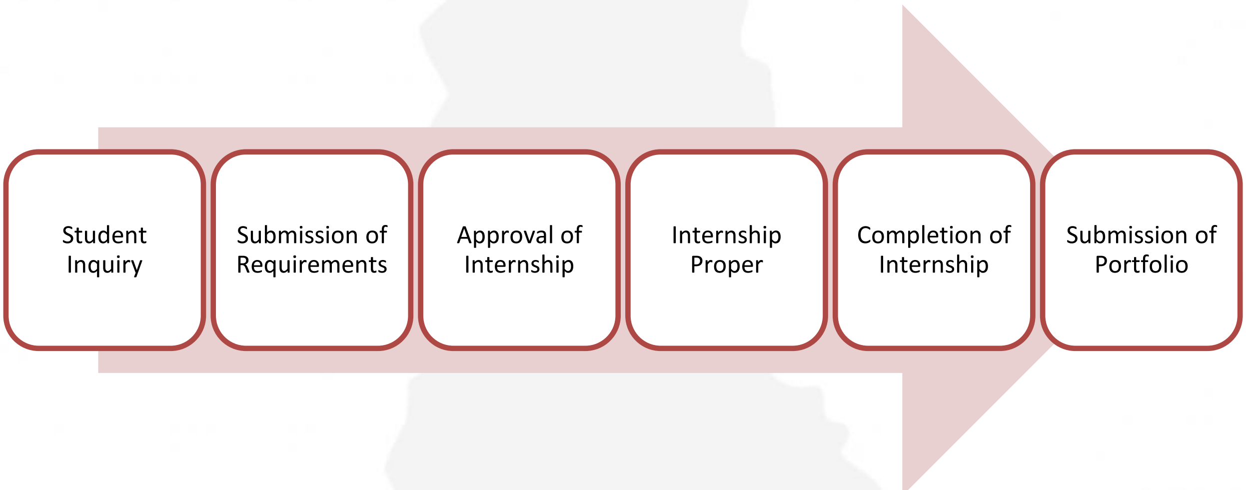


Figure 1: Flowchart of the entire process from application to completion of practicum/internship.

1. Students will research which institutions accept internships and choose one among them to apply for an internship position.
2. Students will collect the required documents by the partner institution (the accepting institution) and submit.

3. The supervisor will submit a fully accomplished and signed acceptance letter form, indicating the approval of the student internship, with some ideal details of what the training program will be. (Letter of Acceptance Form and Attachments)
4. The student intern will fulfill their duties accordingly.
5. Upon completion of the 200-hour training, the supervisor should provide the student with their Certificate of Completion, signed by the supervisor and/or other authorities in the partner institution. Certificate Template is provided by the institution or the OJT-Coordinator if requested.
6. Students will submit their portfolio to the OJT-Coordinator, with a copy of the Completion Certificate attached in the Appendices.

D. Grading System: Supervisor: 70% OJT Coordinator: 30% Total of 100% Passing total percentage: 75%	• **Supervisor will give 70%** of the grade based on the student's performance using the evaluation sheet to be provided at **the last week of the approved training schedule.** *(attached from the letter of approval form)* • **OJT Coordinator will give 30%** of the grade based on the student's submissions (portfolio and daily journal) and **one-time Supervisor-OJT coordinator** meeting (via **online or in-person**) to be scheduled after the **first half** of the **approved training schedule**

E. List of Expectations/ Possible Task to be delivered by the student

For the guidance of the supervisors, below are the list of expectations/possible tasks to be delivered by the students. These are just recommendations and to give the supervisor an overview of what specific duties and responsibilities should the student accomplish throughout the training. Still, the **supervisor can deviate from the list** whenever needed. The supervisor and student **can make use** the table below as a **checklist**.

CLO1	**Demonstrate an understanding of theoretical knowledge on health physics by performing their duties as a student intern on their chosen institution for at least 200 hours**
	Assist in the calibration and maintenance of radiation detection equipment, such as Geiger counters, scintillation detectors, and ionization chambers.
	Participate in the inspection and evaluation of radiation safety programs and practices in various settings, such as hospitals, research laboratories, and industrial facilities.
	Assist in the planning and execution of radiation safety training programs for staff and other personnel.
	Help in the collection and analysis of radiation exposure data, including environmental monitoring data, worker exposure data, and patient exposure data.
	Participate in the design and implementation of radiation protection programs, such as radiation shielding, contamination control, and waste management.

	Assist in the development and maintenance of radiation safety policies, procedures, and regulations, in compliance with national and international standards.
	Help in the preparation and review of safety reports, safety plans, and safety audits related to radiation safety.
	Participate in the investigation of radiation incidents and accidents, including root cause analysis, corrective actions, and follow-up activities.
	Assist in the evaluation and selection of radiation protection equipment and materials, such as shielding materials, personal protective equipment, and dosimetry devices.
	Participate in the research and development of new technologies and techniques in health and medical physics, such as imaging techniques, radiation therapy, and nuclear medicine.
CLO2	**Exemplify professionalism and work ethical standards with different radiation personnel whether in a hospital or an industrial setting**
	Attend safety meetings and training sessions regularly to stay informed and updated on safety policies, regulations, and best practices.
	Observe and follow safety rules and procedures in the workplace, such as wearing personal protective equipment, maintaining good hygiene, and following proper waste management practices.
	Practice good communication skills when working with radiation personnel and other staff, such as listening actively, asking questions, and providing feedback.
	Maintain a positive attitude and a professional demeanor at all times, even in challenging situations, and work collaboratively with others to achieve common goals.
	Respect the confidentiality of sensitive information, such as patient records and proprietary data, and follow data privacy policies and regulations.
	Adhere to ethical standards of conduct and behavior, such as honesty, integrity, and respect for diversity, and report any violations or concerns to the appropriate authorities.
	Demonstrate a commitment to lifelong learning and professional development by seeking out opportunities to acquire new knowledge and skills, attending conferences and workshops, and networking with other professionals in the field.
	Participate in community service activities or outreach programs related to radiation safety and health physics, such as public lectures, seminars, or health fairs.
	Take responsibility for one's own safety and well-being, as well as that of others in the workplace, by reporting any unsafe conditions or incidents promptly and following appropriate protocols for reporting and investigating incidents.
	Maintain accurate and detailed records of work activities, such as experimental data, safety inspections, and incident reports, and communicate results and findings effectively to colleagues, supervisors, and other stakeholders.
CLO3	**Appreciate health physics as a developing field in the country**
	Attend seminars, conferences, and other events related to health physics to learn about the latest developments and trends in the field.
	Conduct research on the current state of health physics in the Philippines, including the challenges and opportunities for growth and development.
	Explore the different applications of health physics in various industries and sectors, such as healthcare, nuclear energy, and radiation safety.

	Interview professionals in the field, such as health physicists, radiologic technologists, nuclear medicine technologists, and radiation safety officers, to gain insights into their work and experiences.
	Engage in discussions with peers and mentors about the role of health physics in promoting public health and safety, and the importance of promoting awareness and education on radiation safety.
	Visit institutions and facilities that use radiation and nuclear technology, such as hospitals, research laboratories, and power plants, to observe how health physics is applied in practice.
	Participate in outreach programs and advocacy campaigns related to health physics, such as radiation safety awareness campaigns, public lectures, or community health programs.
	Learn about the regulations and guidelines governing the use of radiation and nuclear technology in the Philippines, such as the Food and Drug Administration's (FDA) Center for Device Regulation, Radiation Health and Research (CDRRHR) and the Philippine Nuclear Research Institute (PNRI).
	Develop a deeper understanding of the ethical and social implications of using radiation and nuclear technology in society and engage in discussions on how to balance the benefits and risks of these technologies.
	Reflect on one's own experiences and learning and identify areas for further growth and development in the field of health physics.
CLO4	**Create a portfolio summarizing the experience of a health physics intern in the field (hospital, research, or industrial setting)**

F. Training Design

- The supervisor has the option to choose from the following training design:
  A. Full in-person practicum
  B. Hybrid in-person/online practicum (preferably higher percentage of reporting in the field)
- The **full in-person** practicum design is a 100% face-to-face delivery of the training.
- The **hybrid design** allows both the supervisor and the student to have flexible delivery of the training. For example, the student can be in-person reporting from Monday-Wednesday for a certain agreed number of duty hours, and then the rest of the week will be a work-from-home set up, doing tasks that can be done remotely. The online monitoring will be up to the supervisor. This option is also in accordance with the COVID19 protocols of **social distancing.** Preferably, the percentage of in-person training (60%) is higher than the time spent on a work-from-home setup (40%). Again, the supervisor may deviate from this recommendation.

G. Scheduling

By recommendation, the supervisor can choose from the schedule types:

Requirement: **180 hours of training**	No of working hours a **week**	No of working hours a **day**	No of **weeks** for **completion**
Schedule A	40	8	4.5
Schedule B	30	6	6
Schedule C	20	4	9

Note: The remaining 20 hours will be spent by the student in creating their portfolio and other individual submissions to the OJT-coordinator. Only 180 hours will be given as contact time of the student with the supervisor. (180 hours supervisory tasks + 20 hours of OJT-coordinator tasks = 200 hours)

The **supervisor may deviate** from the recommendation above. Please note that some students are enrolled in one or two more subjects aside from their practicum. Hence, an additional one buffer week is requested to finish the required hours of training. Moreover, training **design B** (hybrid training) is suggested to make the time flexible for both students and supervisors, preventing unnecessary extension of the internship. However, if there will be unforeseen circumstances, if the student cannot finish the internship on or before the last day of the midyear term (when the students are enrolled for practicum) the **internship extension may be granted** if there will be a **letter of agreement** from the supervisor and OJT-coordinator. This letter will be the replacement for the completion certificate needed in the portfolio, as a partial requirement of the course.

H. Expected STUDENT output

Requirement	Submitted to	When
Daily Journal	OJT-coordinator	Weekly
Portfolio (with pictures) in PDF format	OJT-coordinator	At the End of the MidYear Term
Feedback form	OJT-coordinator (required) and/or Supervisor (optional)	At the End of the MidYear Term

I. Requested documents from the SUPERVISOR

Requirement	Submitted to	When
Signed Letter of Acceptance Form	OJT-coordinator	Before the internship proper
Signed Certificate of Completion/ Letter of Extension	OJT-coordinator	At the End of the MidYear Term
Accomplished Evaluation Form for the Student	OJT-coordinator (required) and/or Student (optional)	At the End of the Internship Program

Noted by:

Eric M. Inocencio, MSc.
OJT-Coordinator (BS Applied Physics Program)
Department of Physical Sciences and Mathematics, College of Arts and Sciences
University of the Philippines - Manila

Marie Josephine M. De Luna, Ph.D.
Department Chair
Department of Physical Sciences and Mathematics, College of Arts and Sciences
University of the Philippines - Manila

Training Portfolio

2.1 Daily Journals

This portion provides the daily journals accumulate during the internship dated June 22, 2026 to [insert end date here]. The following gives an outline of the main progress done.

Date	Progress
June 22, 2026	Orientation
	B0: Introduction to Paper (Bagunu and Galapon (2025)
June 23, 2026	**B0:** Primary Reading of Whole Paper.
June 24, 2026	**B1:** Start of Analysis and Derivations.
	B1: Research on the Weyl, simplest-Symmetrization, and Born-Jordan ordering rules. Provided short report on progress to the paper to supervisor.
June 25, 2026	**B1:** Derived relationship of the Schrodinger and Heisenberg pictures of quantum mechanics.
June 26, 2026	**B2:** Derivation of the images from the Born-Jordan quantization.
June 29, 2026	**B3:** Continuation of the derivation of the images from the Born-Jordan quantization.
June 30, 2026	**B4:** Derivation of the images as a result of the modifications to the differential quotient of first type.
July 1, 2026	**B5:** Suzuki's analysis on Banach spaces.
	B5: Provision of the rigorous proof for the application of the differential quotient of the second type to operators.
July 2, 2026	**B5:** Derivation of the theorems (1).
July 3, 2026	**B5:** Derivation of the theorems (2).
	B5: Proof for Euler's reflection formula and its relationship to the gamma and beta functions.
July 7, 2026	**B5:** Derivation of the theorems (3).
July 9, 2026	**BF:** Initial report/partial report to Dr. Ralph Ferrales.
July 13, 2026	Exploration of the derivation of the Weyl ordering.
July 14, 2026	Reading of Domingo's paper (2016) on Algebra of Non-Weyl operators.
July 15, 2026	**B6:** Quantization of the Hamilton Equations of Motion
	BA: Exploration of case for the harmonic oscillator.
	Further research on the generalized Weyl transform.
July 16, 2026	**BF:** Completion of the Paper Analysis.
July 17, 2026	**BF:** Report of the Whole Paper.

Table 1: The bold sections refer to sections worked on the paper. Please refer to Appendix 3.8.

2.2 OJT Program Review and Personal Professional Plan

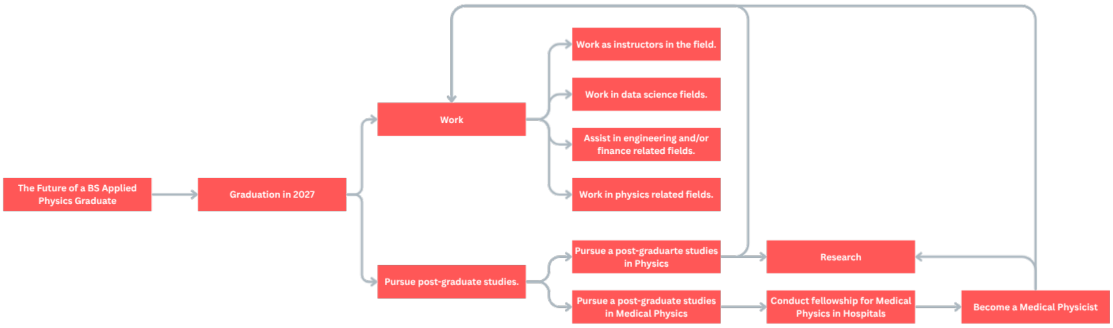
2.2.1 The OJT Program

As a new intern, stepping to the National Institute of Physics at the University of the Philippines Diliman has been a challenge for me. Sure, I have visited the area when I participated in the annual Proceedings of the Samahang Pisika ng Pilipinas conference conducted during the year 2025. However, stepping inside the actual laboratories by which scientific progress in numerous fields of physics was made was amazing to me. The internship started late, June 22, 2026, to be exact, due to the conduct of the 2026 Proceedings of the Samahang Pisika ng Pilipinas conference held at the CAS Annex Building, University of the Philippines Los Banos, of which I presented a poster presentation. The Deputy Director for Research and Extension, Dr. Lim, greeted and welcomed all of us warmly. A tour with the laboratory groups was conducted, inside the restricted portion of the institute. The room I called home for a short time was Room 204, also known as the Theory room, because it housed all of those who wanted to pursue research in Theoretical Physics. It was cold, filled with papers full of scratch equations, and whiteboards plastered over the walls. Fellow interns, Master's students, and PhD students were writing on them, working on a niche problem such that each one admitted that they did not understand each other's work. All except our supervisor, however, Dr. Eric A. Galapon. Originally, we were four, two people from University of the Philippines Manila (Me and Mr. River Villanueva), one from University of the Philippines Los Banos, and one from the Philippine Science High School Cagayan Valley Campus. Together, we spent the time to help each other grow accustomed to the environment we have been exposed with. At the start, we have been assigned papers by Dr. Galapon to study and produce the derivations within the paper. River and I had been assigned Sir Ramon Bagunu's paper "Quantization of the Hamilton Equations of Motion." The main goal was to reproduce the concepts and understand how such research is conducted in the Theory group, specifically the Quantum Physics subgroup. I finished the paper within the day, that is, a preliminary reading where I found the results interesting for the context given.

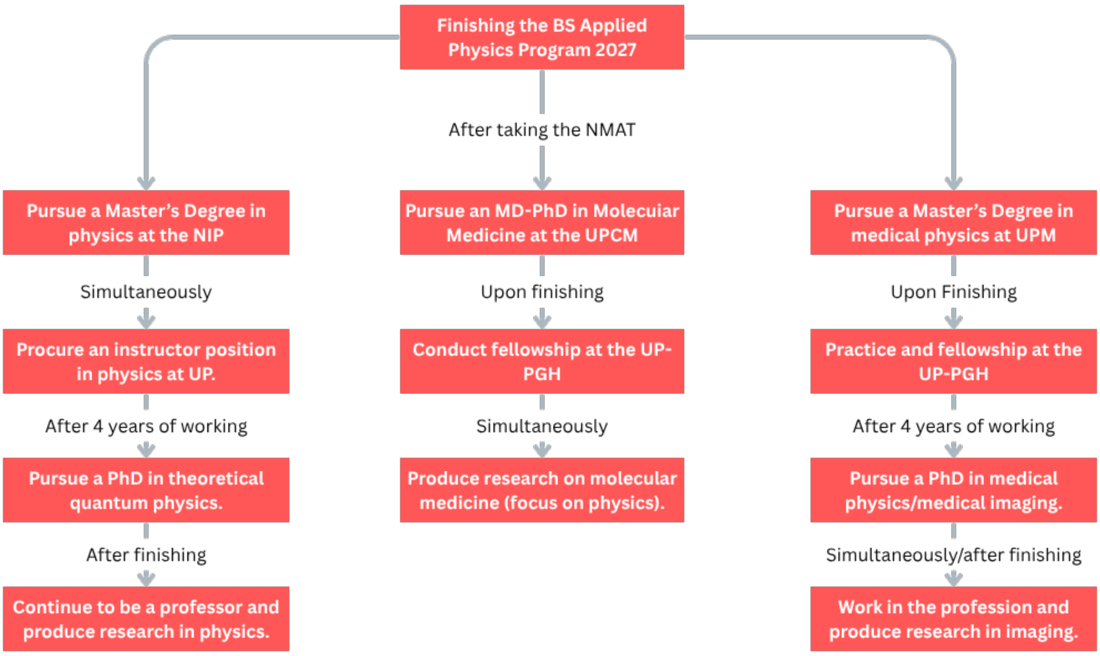
The actual derivations were the hardest part. I had been unfamiliar with the concepts introduced with the paper. Dr. Herbert Domingo did his best to prepare us on the possible concepts we would be exposed with. However, the unfamiliarity made me more excited to work on the paper and understand it but also at the same time be intimidated with it. I struggled with understanding and became insecure as, as compared to my peers who have done significant progress, I was stuck sometimes on some topics wondering "why would this condition be assumed," or "why is this result used." But due to the help of those very own peers and my supervisor who was always to consult upon when prompted, I was able to get past my insecurities and roadblock in the research. I also thought that in Theory, majority of the people would be preoccupied in their work to do anything else. But I soon learned, after taking late night stays in the laboratory due to my part-time job tutoring, that the people here are as sociable as me, lessening my worry that it wouldn't be fun to pursue my further studies in physics.

My time at the Theoretical Physics Group doesn't end here. Temporarily, it does. But I have learned a lot. I learned to ask for help occasionally when I'm unfamiliar or having trouble, as extra perspectives could give significant influences in my own thinking. I may work more on being independent, but it wouldn't hurt to also learn to rely on others to help me.

2.2.2 The Future of a BS Applied Physics Graduate



2.3 Personal Professional Plan



Appendix

3.1 Completion Certificate

Supervisor has not accomplished this yet. This document is to follow.

3.2 Letter of Acceptance

Please see attached file on next page.

NATIONAL INSTITUTE OF PHYSICS
COLLEGE OF SCIENCE
University of the Philippines
Diliman, Quezon City 1101 Metro-Manila
Tel. No. (632) 920-9749 Fax. No. (632) 928-0296

LETTER OF ACCEPTANCE

June 22, 2026

To: **ERIC M. INOCENCIO, MSc.**
OJT-Coordinator – BS Applied Physics Program
Department of Physical Sciences and Mathematics
College of Arts and Sciences
University of the Philippines – Manila

Dear Prof. Inocencio:

This is to inform you that Mr. **Miel Gabriel Lagdan** is accepted and will serve as a student intern in our institution – the **National Institute of Physics**.

Name (Last name, First Name, M.I.)	Student Number
Lagdan, Miel Gabriel	2023-14932

The duration of the internship program will be from June 22, 2026 – July 31, 2026, based on the university calendar; or on any other dates negotiated towards the OJT coordinator. We are glad to have your student be of service in our institution. We hope this will further our relationship as we serve with compassion and excellence.

Sincerely,

ERIC A. GALAPON, PhD.
Professor
National Institute of Physics
College of Science
University of the Philippines - Diliman

3.3 Proposed Training and Design Schedule

3.4 Pictures of Daily Journal

3.5 List of References Read During the Training

MAIN PAPERS:

1. Bagunu, R. J. C., & Galapon, E. A. (2025). *Quantization of the Hamilton equations of motion. Physica Scripta*, *100*(5), 055208. https://doi.org/10.1088/1402-4896/adc047
2. Domingo, H. B., & Galapon, E. A. (2015). *Generalized Weyl transform for operator ordering: Polynomial functions in phase space. Journal of Mathematical Physics*, *56*(2). https://doi.org/10.1063/1.4907561
3. Domingo, H. B. (2016). *Algebra of Non-Weyl Operators and Quantization of Analytic and Non-Analytic Classical Functions.* [Unpublished dissertation]. University of the Philippines Diliman.

RESOURCES:

1. Olsen, P. A., Rennie, S. J., & Goel, V. (2012). *Efficient automatic differentiation of matrix functions.* In *Lecture Notes in Computational Science and Engineering* (pp. 71–81). https://doi.org/10.1007/978-3-642-30023-3_7
2. Muller, F. (1997). *The equivalence myth of quantum mechanics—Part I. Studies in History and Philosophy of Science Part B: Studies in History and Philosophy of Modern Physics*, *28*(1), 35–61. https://doi.org/10.1016/S1355-2198(96)00022-6
3. Cohen, L. (1976). *Quantization problem and variational principle in the phase-space formulation of quantum mechanics. Journal of Mathematical Physics*, *17*(10), 1863–1866. https://doi.org/10.1063/1.522807
4. Cohen, L. (1966). *Generalized phase-space distribution functions. Journal of Mathematical Physics*, *7*(5), 781–786. https://doi.org/10.1063/1.1931206
5. McCoy, N. H. (1932). *On the Function in Quantum Mechanics Which Corresponds to a Given Function in Classical Mechanics. Proceedings of the National Academy of Sciences*, *18*(11), 674–676. https://doi.org/10.1073/pnas.18.11.674
6. Moyal, J. E. (1949). *Quantum mechanics as a statistical theory. Mathematical Proceedings of the Cambridge Philosophical Society*, *45*(1), 99–124. https://doi.org/10.1017/S0305004100000487
7. Zachos, C. (2002). *Deformation Quantization: Quantum Mechanics Lives and Works in Phase-Space. International Journal of Modern Physics A*, *17*(3), 297–316. https://doi.org/10.1142/S0217751X02006079
8. Margenau, H., & Hill, R. N. (1961). *Correlation between Measurements in Quantum Theory. Progress of Theoretical Physics*, *26*(5), 722–738. https://doi.org/10.1143/PTP.26.722
9. Daughaday, H., & Nigam, B. P. (1965). *Function in Quantum Mechanics Which Corresponds to a Given Function in Classical Mechanics. Physical Review*, *139*(5B), B1436–B1442. https://doi.org/10.1103/PhysRev.139.B1436
10. Praxmeyer, L., & Wódkiewicz, K. (2003). *Hydrogen atom in phase space: The Kirkwood–Rihaczek representation. Physical Review A*, *67*(5). https://doi.org/10.1103/PhysRevA.67.054502

11. Kirkwood, J. (1933). *Quantum Statistics of Almost Classical Assemblies. Physical Review*, *44*(1), 31–37. https://doi.org/10.1103/PhysRev.44.31

12. Fan, H., Yuan, H., Cai, G., & Jiang, N. (2011). *Operators' ordering: from Weyl ordering to normal ordering. Science China Physics, Mechanics and Astronomy*, *54*(8), 1394–1397. https://doi.org/10.1007/s11433-011-4404-z

13. Fan, H. (1992). *Weyl ordering quantum mechanical operators by virtue of the IWWP technique. Journal of Physics A: Mathematical and General*, *25*(11), 3443–3447. https://doi.org/10.1088/0305-4470/25/11/043

14. Tang, X., Sun, Y., & Su, H. (2010). *Weyl ordering symbol method for studying Wigner function of the damping field. International Journal of Theoretical Physics*, *49*(9), 2016–2027. https://doi.org/10.1007/s10773-010-0385-3

15. Cohen, L. (2012). *The Weyl Operator and its Generalization.* https://doi.org/10.1007/978-3-0348-0294-9

16. Qian, M., & Xu, P. (1988). *Generalized Weyl transformation. Acta Mathematicae Applicatae Sinica English Series*, *4*(3), 275–288. https://doi.org/10.1007/BF02006225

17. De Gosson, M. A., & Luef, F. (2015). *Born–Jordan pseudodifferential calculus, BOPP operators and deformation quantization. Integral Equations and Operator Theory*, *84*(4), 463–485. https://doi.org/10.1007/s00020-015-2273-y

18. Neudecker, H. (1969). *Some Theorems on Matrix Differentiation with Special Reference to Kronecker Matrix Products. Journal of the American Statistical Association*, *64*(327), 953–963. https://doi.org/10.1080/01621459.1969.10501027

19. Bender, C. M., & Dunne, G. V. (1989). *Exact solutions to operator differential equations. Physical Review D*, *40*(8), 2739.

20. Fujii, K., & Suzuki, T. (2004). *A New Symmetric Expression of Weyl Ordering. Modern Physics Letters A*, *19*(11), 827–840. https://doi.org/10.1142/S021773230401374X

21. Fan, H., & Fan, Y. (2002). *Weyl Ordering for Entangled State Representation. International Journal of Modern Physics A*, *17*(5), 701–708. https://doi.org/10.1142/S0217751X02003257

22. Gelfand, I. M., & Fairlie, D. B. (1991). *The algebra of Weyl symmetrised polynomials and its quantum extension. Communications in Mathematical Physics*, *136*(3), 487–499. https://doi.org/10.1007/BF02099070

23. Bargellini, A. E., & Ritelli, D. (2026). *An elementary approach to Euler's reflection formula and its role in the infinite product of the sine function and the Basel problem. Foundations*, *6*(2), 14. https://doi.org/10.3390/foundations6020014

24. Fedak, W. A., & Prentis, J. J. (2009). *The 1925 Born and Jordan paper "On quantum mechanics". American Journal of Physics*, *77*(2), 128–139. https://doi.org/10.1119/1.3009634

25. Born, M., & Jordan, P. (1925). *Zur Quantenmechanik. The European Physical Journal A*, *34*(1), 858–888. https://doi.org/10.1007/BF01328531

26. Suzuki, M. (1999). *General Formulation of Quantum Analysis. Reviews in Mathematical Physics*, *11*(2), 243–265. https://doi.org/10.1142/S0129055X9900009X

3.6 Abstract of Proposed Thesis

Quantization Ambiguities in Entanglement-Enhanced Interferometric Phase Generators

Miel Gabriel Lagdan, Herbert B. Domingo

Measurements involving photons are classically limited by the shot-noise limit. Entanglement in nonlinear interferometers enables these systems to approach or surpass the Heisenberg limit. However, the quantization of phase generators is plagued by ambiguities arising from different operator-ordering prescriptions (e.g., Weyl, Born-Jordan, and simplest symmetrization). This thesis investigates these ambiguities by deriving and comparing the phase sensitivities of entanglement-enhanced interferometric phase generators constructed under various ordering rules. Implications for quantum metrology in nonlinear media are discussed.

3.7 Resume

Please see attached file on next page.

MIEL GABRIEL LAGDAN

Applied Physics Undergraduate | Theoretical Physics Interests

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PROFESSIONAL SUMMARY

An applied physics undergraduate with a strong foundation in mathematical and theoretical physics, complemented by experience in undergraduate-level instruction and research. Knowledgeable in computational tools such as MATLAB, R, and Python, with research interests in theoretical and mathematical physics, particularly quantum theory and particle physics.

EXPERIENCE

VOLUNTEER TUTOR

University of the Philippines Manila — **October 2023 – Present**

Ermita, Manila, Philippines — PART-TIME

- Provides support to 30+ undergraduate students seeking reinforcement in learning topics involving algebra, calculus, real analysis, theoretical mechanics, electromagnetism, and quantum mechanics.
- Produces numerous helpful resources, including sample examinations, that helped many students in honing their skills in solving problems involving analysis and physics.
- Conducts one-to-one or many-to-one tutorial sessions to the students at the University of the Philippines Manila free of charge and helps other tutors with assistance in procuring resources for their students.

TUTOR

Great Minds Tutorial and Review Center — **August 2025 – Present**

Quezon City, Philippines — PART-TIME

- Provides supplementary help to 10+ elementary and high school students in topics related to mathematics and science (specifically physics) and recommends studying regimens tailored to each student.
- Creates and procures helpful materials in pedagogy useful to the student and the tutoring community.
- Assists students in preparation for upcoming academic competitions, nationally and internationally.

RESEARCH EXPERIENCE

Energies of two-flavor neutrino oscillations in uniform mass distribution using Dirac equation — **2025**

Proceedings of the Samahang Pisika ng Pilipinas

Miel Gabriel Lagdan, Jourdyn Roland N. Garcia, Herbert B. Domingo

- Performed theoretical analysis of neutrino oscillations of particles of uniform masses using coupled Dirac equations.
- Used Elisabetta Sassaroli's model (1999) of neutrino oscillations and extended it by applying small mass increments.
- Provided a framework for uniform mass distribution models, with recommendations for extensions for varying mass.

Assessment of the Effectiveness of "Thermika," a Serious Game, in Teaching Thermochemistry — **2024**

Philippine Federation of Chemistry Societies, Inc.

Miel Gabriel Lagdan, Pauline Khyla D. Gabatin, Niño Sherwin Lumancas, Roma Balagtas-Arguelles, Toshi Akira Capintog

- Development and statistical analysis of a serious game's effectiveness to 84 high school students studying thermochemistry.
- Results demonstrate statistically significant and effective application of intervention with statistical evidence.
- An online, playable version was uploaded to **itch.io**, where students are free to access the game to assist in learning.

EDUCATION

Bachelor of Science in Applied Physics *(Health Physics Concentration)* — Expected Graduation: **2027**

University of the Philippines Manila — **August 2023 – Present**

- With relevant coursework in radiation physics, statistical methods, magnetic resonance imaging, and computational physics.
- Joined a research laboratory in the University of the Philippines Manila called "Laboratory for Applied Mathematical Physics."

Secondary Education

Caloocan City Science High School — **September 2021 – July 2023**

SKILLS

Technical Skills:
Python (numerical analysis)
MATLAB & R (statistical analysis, simulations)
LaTeX (technical writing)
Arduino (C++) (hardware interfacing)

Analytical Skills:
Mathematical modeling of physical systems
Differential equation analysis (ODEs & PDEs)
Approximation techniques
Statistical techniques

Other Skills:
Communication
Teamwork
Adaptability
Leadership

3.8 OUTPUT: Analysis of Paper (Bagunu & Galapon)

Please see linked file to research repository.

Link to Repository: Analysis Paper (Bagunu and Galapon)